CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EE) Examination

November 2013

EE203/EE203.01 Electrical Measurement & Measuring Instruments

Date: 14.11.2013, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain briefly Nalder Lipman power factor meter. [04]

(b) Write a short note on Electrodynamometer type frequency meter. [03]

Q - 2 (a) Draw equivalent circuit and phasor diagram of Potential Transformer. Derive actual transformation ratio for potential transformer. [07]

(b) With change in following parameters explain effect on characteristics of Potential Transformer: [07]

- Change in secondary current or VA
- Change in frequency
- Change in power factor of secondary burden
- Change in primary voltage

OR

Q - 2 (a) Write a short note on mutual inductance method for testing of current transformer. [05]

(b) Write a short note on Varley loop test for localization of cable faults. [05]

(c) A current transformer with a bar primary has 300 turns in its secondary winding. The resistance and reactance of the secondary circuit are 1.5 Ω and 1.0 Ω respectively including the transformer winding. With 5 A flowing in the secondary winding, the magnetizing mmf is 100 A and the iron loss is 1.2 W. Determine the ratio and phase angle errors. [04]

Q - 3 (a) Write a short note on repulsion type moving iron instrument. Also, derive general torque equation for moving iron instrument. [07]

(b) Explain construction and operating principle of electrodynamometer type instrument. State advantages and disadvantages of electrodynamometer type instrument. [07]

OR

Q - 3 (a) Write a short note on quadrant electrometer. State advantages and disadvantages of electrostatic instruments. [06]

(b) State and explain advantages and disadvantages of permanent magnet moving coil type instrument. [04]

(c) Write a short note on megger. [04]

SECTION – II

Q - 4	(a)	Explain relative limiting error.	[03]
	(b)	Explain True value and Precision related to measuring instruments.	[04]
Q - 5	(a)	Explain in detail Ammeter-Voltmeter method and Substitution method for measurement of medium order resistance.	[07]
	(b)	What is the procedure of standardisation for slide wire potentiometer?	[03]
	(c)	A basic slide wire potentiometer has a working battery voltage of 3.0 V with negligible internal resistance. The resistance of slide wire is 400 Ω and its length is 200 cm. A 200 cm scale is placed along the slide wire. The slide wire has 1 mm scale divisions and it is possible to read upto 1/5 of a division. The instrument is standardized with 1.018 V standard cell with sliding contact at the 101.8 cm mark on scale. Calculate • Working current • The resistance of series rheostat • The measurement range • Resolution of the instrument	[04]

OR

Q - 5	(a)	Explain in detail Gall-Tinsley A.C. potentiometer. Also, mention the errors introduced in measurement by using this potentiometer.	[07]
	(b)	Write short note on following methods for measurement of high resistance: • Direct deflection method • Loss of charge method	[07]
Q - 6	(a)	Write a short note on step by step method for determination of B-H curve for ring shaped specimen	[05]
	(b)	Explain operation of Hay's bridge and also derive its balance equations	[05]
	(c)	Explain working of Maxwell's inductance bridge.	[04]

OR

Q - 6	(a)	Explain in detail magnetic potentiometer.	[05]
	(b)	Explain working of Schering bridge and also derive its balance equations.	[05]
	(c)	The arms of a five node bridge are as follows: Arm ab consists of unknown impedance (R_1,L_1) in series with a non-inductive variable resistor r_1. Arm bc consists of resistor R_3 = 100 Ω. Arm cd consists of R_4 = 200 Ω. Arm da consists of R_2 = 250 Ω. Arm de consists of a variable resistor r. Arm ec consists of loss-less capacitor C = 1μF, and arm be consists of a detector. An a.c. supply is connected between a and c. Calculate R_1 and L_1 when under balance conditions r_1 = 43.1 Ω and r = 229.7 Ω	[04]

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